

Pawan Kumar

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Professional Summary

Backend-focused Software Engineer with experience building high-performance APIs, scalable AWS infrastructure, and AI-powered agentic systems in production environments. Skilled in Golang, cloud-native architectures, and LLM-based workflows including RAG, LangChain, and LangGraph to automate debugging, data access, and decision-making.

Education

Indian Institute of Technology (BHU), Varanasi
B.Tech in Electronics Engineering

Nov 2020 – May 2024
GPA: 8.70 / 10

Experience

Software Engineer — Samsung, Delhi
Backend Engineering

Jul 2024 – Present

- Accelerated API performance by **98% (500ms → 10ms)** through query optimization, connection pooling, and indexing.
- Engineered and productionized **8+ REST APIs** supporting **50K+ daily requests** across internal services.
- Architected a centralized **authentication and request validation service**, securing APIs across **6+ microservices**.
- **Tech:** Golang, Iris, Java, Spring Boot, SQL

AWS & Infrastructure

- Optimized AWS infrastructure using **ECS, Lambda, RDS, and 5+ more services** for production workloads.
- Orchestrated PostgreSQL upgrades using **blue-green deployment**, with **0 downtime**.
- Directed AWS account migration via **Terraform**, across environments leads to reduce migration time by 2-3 weeks.
- Streamlined CI/CD workflows with Jenkins, reducing deployment cycles by **80%**.

AI / Agentic Systems

- Conceived and launched **ServiceHub**, an AI-driven internal platform for intelligent service debugging and data access.
- Integrated **Retrieval-Augmented Generation (RAG)** pipelines to answer internal service queries using contextual knowledge of more than 10 GB data.
- Engineered an autonomous **log-analysis agent** that analyzes failed CloudWatch logs using LLM reasoning, reducing manual debugging by **80%**.
- Designed a secure **natural language-to-SQL workflow** with human approval gates across **10+ database tables**.
- Coordinated multi-step reasoning workflows using **LangChain and LangGraph** with tool and vector store integration.
- **Tech:** LLMs, Prompt Engineering, RAG, LangChain, LangGraph, Vector DB, IAM

Projects

Intelligent Data & Analytics Jewellery Platform

GitHub

- Developed a platform combining **AI-driven SQL generation** and **real-time analytics dashboards** for business insights.
- Automated billing, inventory tracking, and transaction analytics, reducing manual operations by **70%+**.
- Improved internal query productivity by **75%** through approval-based execution workflows.

Technologies

- **Languages & Tools:** Go, Python, Java, C++, SQL, JMeter, Postman, Swagger, GitHub
- **AI & Agents:** LLMs, RAG, LangChain, LangGraph, Prompt Engineering
- **Cloud & DevOps:** AWS (EC2, ECS, Lambda, RDS, S3, IAM, API Gateway), Terraform, Docker, Jenkins, CI/CD

Position of Responsibility & Achievements

- **Captain, Volleyball Team** — Represented institute at **INTER IIT Sports Fest 2024**
- **Awards:** Best Performance Award (Aug 2025); Best Growth Project Award (2025)
- **Courses:** Cloud Computing, AWS Skill Builder Labs, Prompt Engineering, Generative AI, Agentic Systems